

ReadMe Document

This ReadMe document will help you with a quick navigation through the contents of the Pre-Reading material, Preview videos, Individual Labs and Mini-Hackathon/Hackathon for the upcoming session

Outcomes

After completion of this week's sessions, & activity participants will be able to :

1. Understand how the image is fed into ML models i.e., different techniques for getting features out of them
2. Understand some properties of sound and techniques for getting features out of them as a spectrogram, MFCC, etc.
3. Understand & implement SVM for nonlinear classification problems and the concept of Kernels-Polynomial, Gaussian(RBF), etc.
4. Understand & implement SVM for nonlinear classification problems and the concept of KernelsPolynomial, Gaussian(RBF), etc.
5. Understand the Role of Humans & their interventions in ML Loops
6. Understand & implement different ensemble methods, viz. bagging, boosting, etc.

Lecture Topics

1. Features for Perception: Image & Audio
2. Non- linear SVM & Kernels
3. Human in the loop, Ensemble Methods

Preview Videos

The Preview Videos Link for the respective sessions are available a week in advance (i.e. every Monday), to help you understand the concepts that will be covered in the upcoming lectures. There could be one or more videos depending on the lecture details. The video/videos should be viewed before attending the lecture for better understanding.

There are 4 preview videos for this session and they should be viewed as given below:

1. Perception: This is a 3-minute video. From this video you will learn about feature representation for text, image and speech
2. Features for Image: This is a 19 mins video. This video helps you to understand 3 generations of representations: Hand-crafted, Statistical and Learned.
3. SVM Nonlinear: This is a 7-minute 42-second video. This video explains about non-linear feature mapping and how to convert nonlinear problems to linear problems.

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4. Human in the Loop Systems: This is a 7-minute video. From this video, you will understand how Human in the Loop system and Relevance Feedback work.
5. Ensemble Methods: This is a 10-minute video. From this video, you will understand how Ensemble Methods work. 1. You will know how to combine different classifiers like KNN, Decision trees, etc. using Ensemble techniques 2. You will learn about Bagging, Boosting and Random Forest Classifier.

Pre Reading Material

The pre-reading material is given on (1) MFCC, (2) Support Vector Machines (SVM), (3) Ensemble Methods,. You should read the material before you attend the lecture. This pre-reading will provide the gist of the lecture and facilitate your understanding during the lecture.

Reference Material

The Reference Material is given on (1) MFCC, (2) Support Vector Machines (SVM) . You should read the material after attending the lecture. This reference material will provide further depth into the topic.

Individual Labs

1. Features for Perception
2. SVM (Non-linear)
3. Ensemble Methods